

EPHEMORA CELL

Same attack. Different boundary.

Eight attack attempts against one container.

Three security boundaries. Measured, not promised.

In plain words: AI-written code and third-party tools are a black box.
We run them — behind a boundary the code cannot loosen.

Docker: 0/8 stopped · Hardened Docker: 2/8 · Ephemora Cell: 8/8

Same image (python:3.12-slim), pinned by digest · measured 2026-09-20

01 — THE BASELINE

A stock container lets it all through

run system commands (id, ls, ...)

● ALLOWED

spawn its own processes

● ALLOWED

send onto the network

● ALLOWED

write straight to the disk

● ALLOWED

read foreign files (/etc/passwd)

● ALLOWED

break out of the cage via symlink

● ALLOWED

jam all CPU cores

● ALLOWED

read the environment (keys, passwords)

● ALLOWED

In plain words: A normal Docker container is a cage without a lock — good for separating, not for blocking. Everything here is the normal case.

Add every flag. The code is still inside.

run system commands (id, ls, ...)

● ALLOWED

spawn its own processes

● ALLOWED

send onto the network

● ALLOWED

write straight to the disk

● BLOCKED

read foreign files (/etc/passwd)

● ALLOWED

break out of the cage via symlink

● BLOCKED

jam all CPU cores

● ALLOWED

read the environment (keys, passwords)

● ALLOWED

In plain words: `--read-only` & friends wall the container off from outside —
the two stops are side effects. The guest itself barely notices.

```
--network none --read-only --cap-drop=ALL --pids-limit 64
```

python benchmarks/hardened_docker_probe.py → 2/8 stopped · 3/6

03 — THE WASM BOUNDARY

These doors simply don't exist

run system commands (id, ls, ...)

● BLOCKED

spawn its own processes

● BLOCKED

send onto the network

● BLOCKED

write straight to the disk

● BLOCKED

read foreign files (/etc/passwd)

● BLOCKED

break out of the cage via symlink

● BLOCKED

jam all CPU cores

● BLOCKED

read the environment (keys, passwords)

● BLOCKED

In plain words: Ephemora Cell runs code in WebAssembly — a form with no network, process or shell access. Not switched off: never present. Checked with a counter-test: what's allowed must work, or the run doesn't count.

04 — WHAT IT COSTS

~0.51 ms

warm, end-to-end, pooled (median)

~**3M** executions/hour per core

~**5.5M**/hour — pooled engine

In plain words: Securing costs almost nothing here — one run is under a millisecond. So *every* call gets secured, not just the risky ones.

128 MB memory · CPU counted by fuel · max 10 KB output · no network.

Every run returns a record: status · fuel · time · rules.

VERIFY IT

Verifying. Not claimed.

```
$ pip install ephemora-cell  
$ python benchmarks/verify_8_vectors.py → 8/8 stopped  
$ python benchmarks/hardened_docker_probe.py → 2/8  
stopped
```

In plain words: Every number in this deck is a committed measurement file in the repo:
dated and reproducible. No trust required.